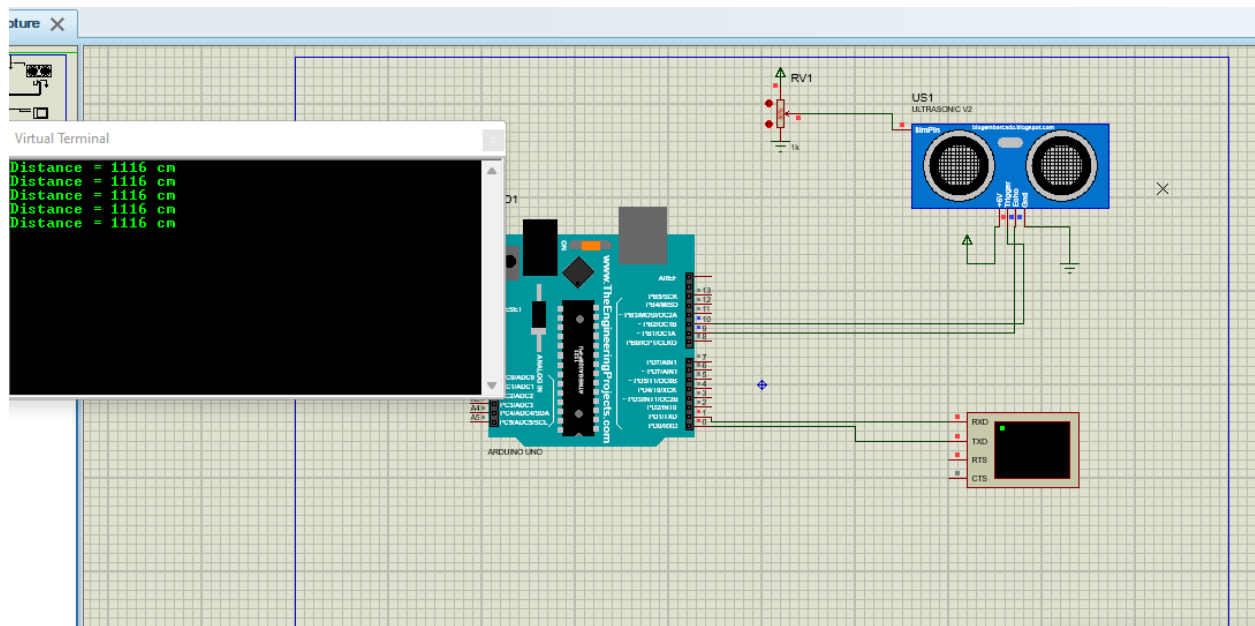


Embedded systems exp 9(arduino_ULTRASONIC)



Program

```
int trig = 10;
int echo = 9;
long duration;
int cm;

void setup()
{
    pinMode(trig, OUTPUT);
    pinMode(echo, INPUT);
    Serial.begin(9600);
}

void loop()
{
    // Clear trigger
    digitalWrite(trig, LOW);
    delayMicroseconds(2);

    // Send 10 µs pulse
    digitalWrite(trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(trig, LOW);

    // Read echo time
```

```
duration = pulseIn(echo, HIGH);

// Calculate distance in cm
cm = duration * 0.0343 / 2;

Serial.print("Distance = ");
Serial.print(cm);
Serial.println(" cm");

delay(500);
}
```